NUTRITION

Goo

📝 Part 1: In-Chapter Point-Wise Notes 🍎🥦

🎯 11.1 Need of Nutrition & Key Terms

- **Nutrition:** The supply of essential organic and inorganic chemical compounds (nutrients) to the body.
- **Six Main Purposes of Food:**
    1. **Growth:** Building up new protoplasm or cells.
    2. **Repair:** Providing material for worn-out or injured cells.
    3. **Energy:** Providing energy needed by the body to carry out life functions.
    4. **Maintenance:** Maintaining the chemical composition of cells.
    5. **Provision of raw materials:** For the manufacture of various secretions (enzymes, hormones, sweat, milk).
    6. **Protection:** From disease and infection.
- **Diet vs. Balanced Diet:** Diet is the combination of food items eaten in meals. A **balanced diet** contains all principal constituents of food in proper quantity. It depends on four factors: Age, Sex, Lifestyle/nature of work, and Special needs (sick person, pregnant lady, nursing mother).
- **Appetite vs. Hunger:** Appetite is the *desire* to eat, while hunger is the *uneasy sensation* due to a lack of food.

📊 11.2 Classes of Nutrients (Food Substances)

There are six classes of food substances: Carbohydrates, fats, proteins, mineral salts, vitamins, and water.

1. CARBOHYDRATES (Principal Energy Sources) 🍞

- Compounds of Carbon (C), Hydrogen (H), and Oxygen (O).

- **Crucial Ratio:** Hydrogen and oxygen are always present in the ratio of **2:1**.
- They are oxidised in the cells to release energy.
- **A. Sugars (Soluble & Sweet):**
    - **Monosaccharides (Single/Simple Sugars):** Formula $C_6H_{12}O_6$. Require no digestion and are straightaway absorbed into the body.
        - (i) **Glucose:** Grape sugar; the most common and simplest sugar. *(Note: "Blood sugar" specifically means the glucose level in blood. 1 mole of glucose releases 686 kcal of energy).*
        - (ii) **Fructose:** Common in plants; fruit sugar.
        - (iii) **Galactose:** Simple sugar found in milk.
    - **Disaccharides (Double Sugars):** Formula $C_{12}H_{22}O_{11}$. Require digestion to get absorbed.
        - (i) **Sucrose:** Commercial sugar (from sugarcane/sugar-beet). Made of glucose + fructose.
        - (ii) **Maltose:** Malt sugar. Made of two glucose molecules.
        - (iii) **Lactose:** Milk sugar. Made of glucose + galactose.
- **B. Polysaccharides (Insoluble Carbohydrates):**
    - **Starch:** Formula $(C_6H_{10}O_5)_n$. The form in which plants store carbohydrates (potatoes, wheat, grains).
    - **Glycogen:** The form in which carbohydrates are stored in animals (majorly in the liver and muscles).
    - **Cellulose:** Found in plant cell walls. It is not used/digested by our body, but it acts as **Roughage**.

🌾 **The Role of Roughage (Non-digestible cellulose)**

- Obtained from salads, green vegetables, fruits, and outer coats of grains (like "chokar" / Dalia / cabbage).

- **Functions:** Being fibrous, it absorbs and retains a lot of water. It helps keep the faecal matter soft, prevents constipation, stimulates muscle contraction in the intestinal wall, and helps in the removal of cholesterol and toxins.

2. FATS 🧈

- Composed of C, H, and O (but the oxygen content is much less than in carbs).
- Made up of **fatty acids and glycerol**.
- **Energy Powerhouse:** They produce **more energy** than carbohydrates (1 mole of fat releases 9.45 kcal of energy).
- **Functions:** Produces energy, acts as an important storage form of food, serves as a solvent for fat-soluble vitamins, and fat under the skin protects the body against rapid loss of heat (insulation).

3. PROTEINS 🥚🥩

- Chemical molecules containing C, H, O, and **Nitrogen (N)**. Some may also contain sulphur and phosphorus.
- **Amino acids** are the simple, smaller units of proteins.
- **Functions:** Provide chemical material for the **growth and repair** of body cells and tissues. In emergencies, they may also be oxidised to release energy.

⚠️ Protein-Energy Malnutrition (Deficiency Diseases)

Deficiency of proteins and energy-giving foods causes two severe diseases in children:

- **(i) Kwashiorkor (Severe Protein Deficiency)**
    - **Cause:** Affects young children (1-5 years) when mothers stop breast-feeding early and the child is given a diet strictly poor in proteins (usually just starchy carbs).
    - *Name Origin:* An African term meaning "the sickness the child gets when a new baby comes" (as breast milk is no longer available).

    - **Symptoms:** Underweight, **belly protruding out**, skin getting dark and scaly, stunted growth, loss of appetite, repeated diarrhoea, enlarged liver, and **Oedema** (swelling of feet and face due to water accumulation in tissues).
    - **Treatment:** A protein-rich diet (pulses, milk, egg, fish, meat).
- **(ii) Marasmus (Deficiency of Carbs, Fats, AND Proteins)**
    - **Cause:** Affects infants usually below 1 year. Caused by sudden stoppage of breast feeding followed by a diet poor in **energy-giving foods (carbs/fats) and proteins**.
    - **Symptoms:** Less body weight, severe degeneration (body looks as if formed of muscles, skin, and bones only), skin becomes loosely folded, thin face/limbs, retarded physical/mental growth, prominent ribs. *(Unlike Kwashiorkor, there is NO oedema and NO darkening of the skin).*
    - **Treatment:** A diet rich in both proteins and carbohydrates.

4. MINERAL SALTS 🧂

Needed in small quantities for vital functions:

- **Calcium and Phosphorus:** Strengthen bones and teeth. Calcium is required for **blood clotting**. Phosphorus is required for chemical processes like ATP (energy) production in cell respiration.
- **Iron:** Has a major role in the formation of **haemoglobin**; deficiency leads to anaemia.
- **Iodine:** Required for proper working of the **thyroid**; deficiency leads to goitre.
- **Potassium and Sodium:** Needed for cell permeability, especially in **nerve cells**.

5. VITAMINS 💊

- Chemical substances needed in minute amounts to maintain a healthy body. Most act as **catalysts or enzymes** in essential cell metabolism.
- **Fat-soluble Vitamins (A, D, E, K):** Can be stored in the body for a longer period of time.
- **Water-soluble Vitamins (B-complex and C):** Cannot be stored for a long period of time; excess water-soluble vitamins are easily excreted out.
- *Note on Absences:* Absence or shortage over a continued period causes specific **deficiency diseases**. *(Note: Per your handwritten annotation crossing out the detailed Table 11.2, that specific detailed list is officially ignored for this study section!)*

📝 Part 1: In-Chapter Point-Wise Notes 💧🥗

💧 6. WATER (The Indispensable Nutrient)

- **Body Composition:** Water is absolutely indispensable, making up about **two-thirds (2/3)** of our total body weight.
- **Key Functions in the Body:**
    1. **Solvent:** It acts as a universal solvent in the body for thousands of organic and inorganic substances.
    2. **Digestion:** It is used to produce vital digestive juices.
    3. **Transportation:** It helps in the transportation of digested foods and oxygen throughout the body.
    4. **Excretion:** It is strictly required for the excretion of soluble wastes (like urine).
    5. **Temperature Regulation:** It is heavily involved in the maintenance of body temperature (via sweating).
- **Loss and Replacement:** Water is regularly lost from the body through sweat, urine, and as water vapour in breath. Therefore, it must be constantly replaced.

- **Daily Requirement:** While the liquids we drink and the food we eat (fruits/vegetables) supply water, we should still actively drink about **7-8 glasses of water every day**.
- *(Note: The detailed table regarding essential vitamins and deficiency diseases has been officially excluded from the syllabus).*

⚙️ An Overall View of the Role of Nutrients

Based on their primary physiological functions, nutrients can be grouped as follows:

- **Energy Providers (Oxidation & ATP):** Carbohydrates & Fats.
- **Builders & Producers (Cells, tissues, hormones, enzymes):** Proteins, Water, & Mineral salts.
- **Regulators (Managing body processes):** Vitamins, Water, & Mineral salts.

⚖️ 11.3 Balanced Diet & Calories

- **Definition:** A balanced diet is one which contains **all** the principal constituents of food in **proper quantity** for a particular person.
- **The Calorie Factor:** One special aspect of a balanced diet is ensuring it provides a sufficient number of calories.
    - **Calorie Definition:** The amount of heat required to raise the temperature of one gram of water by one degree Celsius.
    - **Kilocalorie (kcal):** When considering the energy value of foods, the term "Kilocalorie" is usually used. 1 Kilocalorie = 1000 times the unit calorie.
- **Consequence of Calorie Deficit:** If the daily intake of food does not provide sufficient calories, a person will rapidly lose weight (such as during fasting). The existing food stores of the body are oxidized, and the capacity for physical work is severely reduced.
- *(Note: The specific chart detailing exact calorie requirements for different age groups/professions has been marked to be left out/ignored for this section).*

⚠️ Malnutrition vs. Undernutrition (Crucial Difference)

- **Malnutrition:** This is a broader condition in which a person suffers due to an unbalanced diet. It crucially includes **both deficiency as well as an excess** of nutrients in a person's diet.
- **Undernutrition:** This refers specifically to the condition where a person suffers from *poor nutrition* caused strictly by **not having enough food** containing essential nutrients necessary for growth and health.
- **The Relationship:** Undernutrition is just one part (a subset) of overall malnutrition.

🪨 Essential Mineral Elements in Human Body (Table 11.2)

🟩 I. Macronutrients (Needed in large quantity)

- **1. Calcium** 🥛
  - **Source:** Dairy foods, beans, cabbage.
  - **Function:** Constituent of bones and enamel. Required for muscle contraction, clotting of blood.
  - **Deficiency Disease:** Rickets, poor skeletal growth.
- **2. Sodium** 🧂
  - **Source:** Table salt.
  - **Function:** Regulates acid-base equilibrium.
  - **Deficiency Disease:** Muscular cramps, nerve impulses do not get transmitted.
- **3. Potassium** 🍌
  - **Source:** Banana, potato, citrus fruits.
  - **Function:** Nerve and muscle activity, fluid balance, secretion of neurotransmitter (acetylcholine).
  - **Deficiency Disease:** Nerve impulses do not get transmitted.

- **4. Phosphorus** 🥚
    - **Source:** Dairy products, cereals, beans, nuts, meat, egg.
    - **Function:** Synthesis of nucleic acid, ATP, NADP, bones, enamel, for muscle contraction, conduction of nerve impulse.
    - **Deficiency Disease:** Soft bones.
- **5. Magnesium** 🥬
    - **Source:** Vegetables, whole grain, nuts, sea food, meat.
    - **Function:** Bone and tooth structure, synthesis of enzymes.
    - **Deficiency Disease:** Increased irritability of nervous system.
- **6. Sulphur** 🧅
    - **Source:** Dairy products, onion, garlic, radish, meat, egg.
    - **Function:** Component of protein and co-enzymes.
    - **Deficiency Disease:** Skin problems.
- **7. Chlorine** 🧂
    - **Source:** Table salt.
    - **Function:** Water balance, secretion of HCl in gastric juice.
    - **Deficiency Disease:** Muscular cramps.

🟦 **II. Micronutrients (Needed in small quantity)**

- **1. Iron** 🥩
    - **Source:** Whole cereals, fish, nut, egg yolk, liver, kidney.
    - **Function:** Synthesis of haemoglobin associated with oxidation, reduction reactions.
    - **Deficiency Disease:** Anaemia (deficiency of haemoglobin in blood).
- **2. Cobalt *(Handwritten note: cobalamine)*** 🥛

    - **Source:** Milk, cereals, pulses, liver, red meat.
    - **Function:** Development of red blood cells, component of vitamin B_{12}.
    - **Deficiency Disease:** Pernicious anaemia.
- **3. Copper** 🥜
    - **Source:** Nuts, legumes, liver, kidney.
    - **Function:** Production of melanin, cell oxidation, synthesis of enzymes.
    - **Deficiency Disease:** Loss of body weight, anaemia. *(Handwritten pointer: ➡️ **no melanin**).*
- **4. Zinc** 🍗
    - **Source:** Cereals, pulses, nuts, meat, liver oil.
    - **Function:** Needed for synthesis of enzymes, carbon dioxide transport in blood.
    - **Deficiency Disease:** Retarded growth, skin lesions, albinism.
- **5. Iodine** 🌊
    - **Source:** Iodised salt, water, sea foods.
    - **Function:** Component of thyroxin hormone secreted by thyroid gland.
    - **Deficiency Disease:** Goitre, **cretinism** in children (poor mental and physical growth).
- **6. Fluorine *(Handwritten note: imp.)*** 🚰
    - **Source:** Water, milk.
    - **Function:** Component of tooth enamel and bone.
    - **Deficiency Disease:** Dental decay.
- **7. Manganese** 🥦
    - **Source:** Vegetables.

- **Function:** Bone development, nitrogen-metabolism.
- **Deficiency Disease:** Poor bone development.

- **8. Molybdenum** 🍛
  - **Source:** Pulses, cereals, meat.
  - **Function:** Nitrate assimilation.
  - **Deficiency Disease:** Slight retardation of growth.

💊 Essential Vitamins in Human Body (Table 11.3)

*(Handwritten Mnemonic at top of page: **"He Steals Rickshaw at Night sky"**)* 🚲✨

🟧 (a) FAT-SOLUBLE VITAMINS

- **Vitamin A (Retinol)** 🥕
  - **Rich Sources:** Butter, egg yolk, liver, milk, carrots, leafy green vegetables, yellow fruits, fish liver oils.
  - **Chief Functions:** Promotes growth, resists infection of the skin and mucous membrane, component of visual purple in the retinal cells of the eyes for perception of image.
  - **Deficiency Diseases: Night-blindness** (poor vision in dim light), **Xerophthalmia** (dryness of the cornea and ulceration).
- **Vitamin D (Calciferol)** 🌞
  - **Rich Sources:** Fish liver oils, milk, eggs. Produced by ultraviolet rays in the skin.
  - **Chief Functions:** Helps the body to use calcium and phosphorus to form the bones and the teeth.
  - **Deficiency Diseases: Rickets** in children (bones lack calcium and turn soft). **Osteomalacia** in adults, fractures may occur.
- **Vitamin E (Tocopherol) *(Handwritten note: anti-oxi)*** 🌾
  - **Rich Sources:** Meat, milk, whole wheat.

- **Chief Functions:** Prevents oxidation of vitamin A.
- **Deficiency Diseases:** Muscle weakness, nerve damage, immune system problems.

- **Vitamin K (Phylloquinone)** 🥦
  - **Rich Sources:** Leafy green vegetables, specially cabbage and spinach. Also synthesised in intestines.
  - **Chief Functions:** Needed in clotting of the blood.
  - **Deficiency Diseases:** Haemorrhage (delay in clotting time).

🟦 (b) WATER-SOLUBLE VITAMINS

(Handwritten structural note: B_4, B_8 × meaning they are skipped!) ❌

(Handwritten B-Complex Mnemonic: ***The race Nicely Professionally Performed by crew****)* 🏃 🍽️

- **Vitamin B_1 (Thiamine)** ***(Mnemonic: The)*** 🍞
  - **Rich Sources:** Whole grains, yeast, liver, eggs and lean meat.
  - **Chief Functions:** Increases growth and appetite, helps in digestion and functioning of nervous system.
  - **Deficiency Diseases: Beriberi** (a disease marked by inflammatory or degenerative changes of the nerves, muscles become weak).
- **Vitamin B_2 (Riboflavin)** ***(Mnemonic: race)*** 🥚
  - **Rich Sources:** Eggs, liver, milk, yeast, green vegetables.
  - **Chief Functions:** Regulates oxidation of food.
  - **Deficiency Diseases:** Irritation in eyes and the skin and intestinal disorders, inflammation of the tongue.
- **Vitamin B_3 (Niacin)** ***(Mnemonic: Nicely)*** 🥜
  - **Rich Sources:** Lean meat, liver, milk, eggs, groundnuts, whole grains.

    - **Chief Functions:** Promotes health of the skin and the nervous system.
    - **Deficiency Diseases: Pellagra**, Dermatitis (skin inflammation), loss of memory, diarrhoea, skin lesions, rashes.
- **Vitamin B_5 (Pantothenic acid) *(Mnemonic: Professionally)*** 🥑
    - **Rich Sources:** Mushrooms, avocados, sweet potatoes, lentils.
    - **Chief Functions:** Synthesis and metabolism of fats and proteins, promotes healthy skin, hair, eyes and liver.
    - **Deficiency Diseases:** Fatigue, loss of coordination.
- **Vitamin B_6 (Pyridoxine) *(Mnemonic: Performed)*** 🐟
    - **Rich Sources:** Meat, fish, eggs, cereal bran.
    - **Chief Functions:** Interconversion of amino acids.
    - **Deficiency Diseases:** Skin problems, nerve disorders.
- **Vitamin B_9 (Folic acid) *(Mnemonic: by)*** 🥬
    - **Rich Sources:** Liver, leafy vegetables.
    - **Chief Functions:** Synthesis of haemoglobin.
    - **Deficiency Diseases:** Anaemia (blood deficient in red blood cells, haemoglobin low in total volume).
- **Vitamin B_{12} (Cobalamine) *(Mnemonic: crew)*** 🥩
    - *(Margin note: Cyanocobalamine is its commercial form).*
    - **Rich Sources:** Liver, meat, milk, cereals, pulses.
    - **Chief Functions:** Normal functioning of red blood cells.
    - **Deficiency Diseases: Pernicious anaemia** (a severe anaemia coupled with great weakness and gastrointestinal disturbance, degeneration of spinal cord).
- **Vitamin C (Ascorbic acid)** 🍋

- **Rich Sources:** Fresh citrus fruit (lemon, orange, grape-fruit), tomatoes, germinating seeds. *(Note: It gets readily destroyed by heat).*
- **Chief Functions:** Promotes functioning of capillary walls.
- **Deficiency Diseases:** Increased susceptibility to infections. **Scurvy** (spongy and bleeding gums, loosening of teeth).

🏆 Part 2: End-of-Chapter Master Lists & Exam Cheat Sheets 🧠

📚 1. The Ultimate Glossary (Term vs. Definition)

- **Nutrition:** The supply of essential organic and inorganic chemical compounds (proteins, vitamins, salts, etc.) to the body.
- **Nutrients:** The specific chemical compounds inside food (divided into six classes: Carbs, Fats, Proteins, Minerals, Vitamins, Water).
- **Food:** Any substance we eat or drink which contains these nutrients.
- **Diet:** The combination of food items which we eat in our meals.
- **Balanced Diet:** A diet containing *all* the principal constituents of food in *proper quantity* for a particular person (depends on age, sex, lifestyle, and special needs).
- **Meal:** The food taken at one time to satisfy the appetite.
- **Appetite:** The psychological *desire* to eat.
- **Hunger:** The physical, *uneasy sensation* due to a lack of food.
- **Roughage:** Non-digestible cellulose (fibrous material) that absorbs water, prevents constipation, and helps move undigested food through the intestines.
- **Calorie:** The amount of heat required to raise the temperature of 1 gram of water by 1°C. *(Note: Dieticians usually use Kilocalories / kcal, which is 1000 times a basic calorie).*

- **Malnutrition:** Suffering due to an unbalanced diet (includes *both* deficiency and excess of nutrients).
- **Undernutrition:** A specific type of malnutrition caused strictly by not having *enough* food/nutrients for growth.

⚖️ 2. The "Must-Know" Differences (Directly from Table 11.1)

Examiners love testing the difference between the two protein-energy malnutrition diseases. Here is the perfect breakdown:

Feature 🔍	**Kwashiorkor (Protein Deficiency)** 🛑	**Marasmus (Protein + Carb/Fat Deficiency)** 💀
Age Group Affected	Children between **1-5 years** of age.	Infants usually **up to 1 year** of age.
Root Cause	Deficiency of **proteins** in the diet (mother stops breast-feeding and gives only a starchy diet).	Deficiency of **carbohydrates, proteins, and fats** in the diet (sudden stoppage of breast feeding).
Oedema (Swelling)	**Present:** Oedema of legs and face due to water accumulation.	**Absent:** No oedema whatsoever.
Skin Condition	Skin of the patient gets **dark and scaly**.	**No darkening** of the skin; skin becomes loosely folded.
Muscle Condition	**No severe degeneration** of the muscles.	**Severe degeneration** of muscles; body becomes very weak (thin face, prominent ribs).

Feature 🔍	Kwashiorkor (Protein Deficiency) 🛑	Marasmus (Protein + Carb/Fat Deficiency) 💀
Treatment Diet	Needs a **protein-rich** diet (pulses, milk, egg, meat).	Needs a diet rich in **both proteins and carbohydrates**.

🧪 3. Biochemical Master Matrix: Carbohydrates Breakdown

Carbohydrates are the principal energy sources. Note the strict Carbon, Hydrogen, Oxygen composition where H and O are always in a **2:1 ratio**.

- **A. Monosaccharides (Simple Sugars) ➡️ Formula:** $C_6H_{12}O_6$
  - *Trait:* Require NO digestion; absorbed straightaway.
  - **Glucose:** Grape sugar / Blood sugar. *(Energy Fact: 1 mole of glucose releases exactly 686 kilocalories of energy).*
  - **Fructose:** Fruit sugar (common in plants).
  - **Galactose:** Found in milk.
- **B. Disaccharides (Double Sugars) ➡️ Formula:** $C_{12}H_{22}O_{11}$
  - *Trait:* Made of two simple sugars; require digestion to be absorbed.
  - **Sucrose:** Commercial sugar (Sugarcane/Beet). = *Glucose + Fructose*
  - **Maltose:** Malt sugar. = *Glucose + Glucose*
  - **Lactose:** Milk sugar. = *Glucose + Galactose*
- **C. Polysaccharides (Insoluble Carbs)**
  - *Trait:* Long chains.
  - **Starch:** Formula $(C_6H_{10}O_5)_n$. How plants store carbs (potatoes, grains).
  - **Glycogen:** How animals store carbs (in liver and muscles).

    - **Cellulose:** Plant cell walls. Cannot be digested by humans, acts as Roughage!

⚡ 4. Energy & Physiology Facts (The "Did You Know?" Section)

- **The Fat Energy Rule:** While carbohydrates are the principal energy source, Fats produce *more* energy per gram. **One gram of fat releases 9.45 kcal of energy** (compared to carbs).
- **The Nitrogen Rule:** Carbohydrates and Fats only contain C, H, and O. **Proteins** are unique because they possess **Nitrogen** (and sometimes Sulphur and Phosphorus). Amino acids are the building blocks.
- **Water's Dual Role:** Water provides absolutely ZERO energy/calories, yet you will die without it because it is the universal solvent required for every single chemical reaction in the body and makes up **two-thirds** of your weight.
- **Ignored Tables Note:** As per your handwritten markings (crossing out with a downward arrow and the word *"Leave"*), the detailed tables on Page 120 (Table 11.4 Adult Menu & Calorie Requirements of Different People) are officially skipped for your prep!

🌟 5. Spotlight: Your Handwritten B-Complex Mnemonic!

To remember the sequence and names of the B-Complex water-soluble vitamins, use your brilliant handwritten margin mnemonic:

"The race Nicely Professionally Performed by crew" 🏃 🍽️

- **The** = **Thiamine** (Vitamin B_1) ➡️ *Deficiency: Beriberi*
- **race** = **Riboflavin** (Vitamin B_2)
- **Nicely** = **Niacin** (Vitamin B_3) ➡️ *Deficiency: Pellagra*
- *(Skip B_4)*
- **Professionally** = **Pantothenic acid** (Vitamin B_5)
- **Performed** = **Pyridoxine** (Vitamin B_6)
- *(Skip B_7, B_8)*

- **by** = **Folic acid** (Vitamin B_9)
- **crew** = **Cobalamine** (Vitamin B_{12}) ➡️ *Deficiency: Pernicious anaemia*

*(Note: Your other mnemonic at the top of the page **"He Steals Rickshaw at Night sky"** acts as a great visual trigger for the fat-soluble ones: A, D, E, K!)* 🚲✨